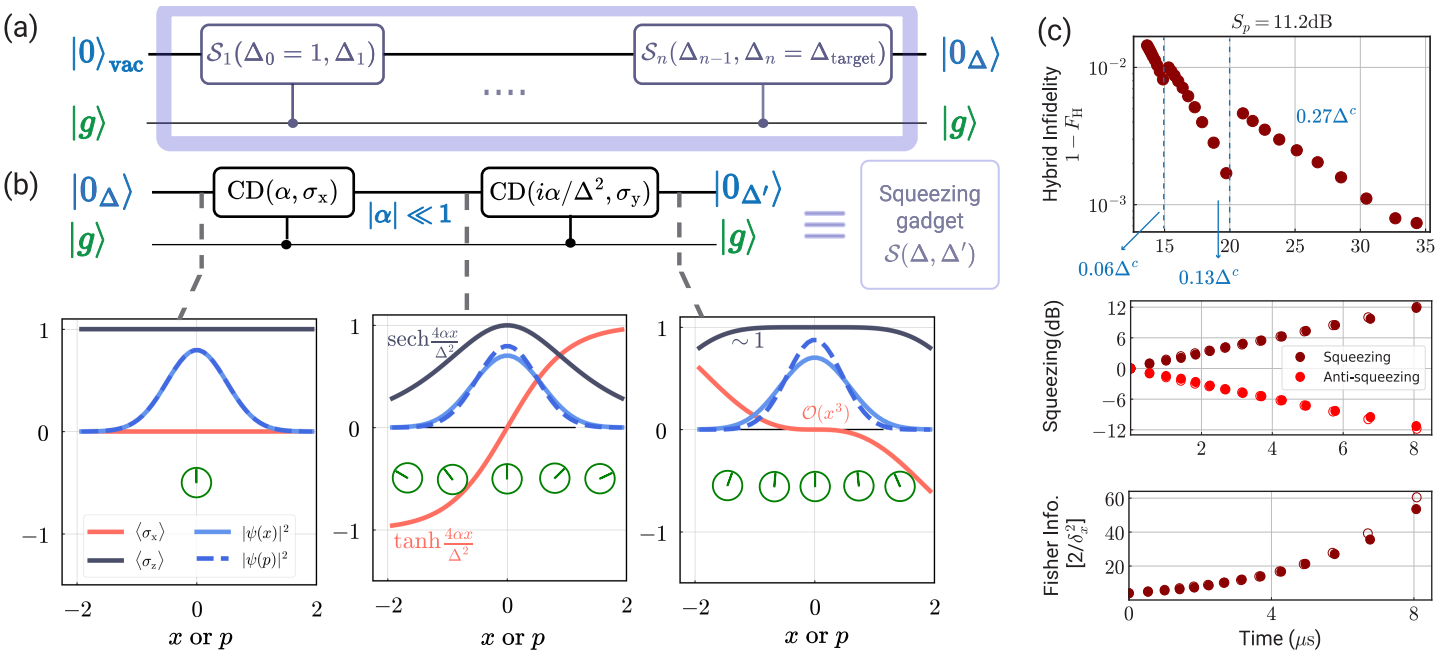

(a)
$|0\rangle_{\rm vac}$
$\mathcal{S}_1(\Delta_0=1,\Delta_1)$
$\mathcal{S}_n(\Delta_{n-1},\Delta_n=\Delta_{\rm target})$
$|0_\Delta\rangle$
$|g\rangle$
$|g\rangle$
(b)
$|0_\Delta\rangle$
$\mathrm{CD}(\alpha,\sigma_{\rm x})$
$|\alpha|\ll 1$
$\mathrm{CD}(i\alpha/\Delta^2,\sigma_{\rm y})$
$|0_{\Delta'}\rangle$
$|g\rangle$
$|g\rangle$
Squeezing gadget
$\mathcal{S}(\Delta,\Delta')$
$\langle\sigma_{\rm x}\rangle$
$\langle\sigma_z\rangle$
$|\psi(x)|^2$
$|\psi(p)|^2$
$\mathrm{sech}\frac{4\alpha x}{\Delta^2}$
$\tanh\frac{4\alpha x}{\Delta^2}$
~ 1
$\mathcal{O}(x^3)$
x or p
x or p
x or p
(c)
$S_p=11.2$dB
Hybrid Infidelity $1-F_{\rm H}$
$0.27\Delta^c$
$0.06\Delta^c$
$0.13\Delta^c$
Squeezing(dB)
Squeezing
Anti-squeezing
Fisher Info. $[2/\delta_x^2]$
Time (μs)